



Indian Institute of Information Technology, Surat
Department of Computer Science & Engineering
Mid Semester Examination, 4th Sem, 2024-25
System Software (CS 403)

Timing: 1:30 PM to 3:00 PM

Date: 05/03/2025

Max Marks: 30

- Note: 1. Attempt all parts of a question at one place.
2. All questions are mandatory
3. Make a necessary assumptions wherever required

Q.1 Answer the following

1. Construct LL (1) parsing table for this grammar. Check whether the input string **abcdef** is accepted or not. [3]
 $S \rightarrow AB \mid eDa$
 $A \rightarrow ab \mid c$
 $B \rightarrow dC$
 $C \rightarrow eC \mid \epsilon$
 $D \rightarrow fD \mid \epsilon$
2. Check if the following grammar is SLR (1) or not. Design parsing table for SLR (1). [3]
 $S \rightarrow S$
 $S \rightarrow \langle L \rangle \mid id$
 $L \rightarrow L, S \mid S$
3. Apply Leftmost and Rightmost derivation and derivation tree for the string **0100110**. [4]
 $S \rightarrow 0S \mid 1AA$
 $A \rightarrow 0 \mid 1A \mid 0B$
 $B \rightarrow 1 \mid 0BB$

Q.2 Answer the following

1. Construct the DAG for the given expression: $x = (a/10 + (b-10)) * (a/10 + (b-10))$ [3]
Count the number of nodes in DAG.
2. Generate Three address code for the following code segment: [3]
main()
{ int a=1; int b[10];
 while(a<=10)
 b[a]=2*a; }
3. Write 3-address code for the given expression and apply methods of 3-address on that. [4]
 $E = ((x+y) - ((x+y) * (x-y))) + ((x+y) * (x-y))$

Q.3 Write a short note

1. Explain Recursive Descent Parser for a given grammar. [3]
 $S \rightarrow aB \mid bA$
 $A \rightarrow c \mid \epsilon$
 $B \rightarrow d \mid \epsilon$
2. Trace the output after each phase of the compiler after assignment statement $a = (b+d) + (c*10)$, if variables are of *int* type. [5]
3. Discuss steps to design the operator precedence table. [2]

****End****